MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular Examination November 2025
First Semester
Skill Development Components (SDC) – MG1SDCSNP100 – PRINCIPLES OF FOOD SCIENCE AND FOOD MICROBIOLOGY
(2025 Admission Onwards)

Duration: 1 hour 30 minutes **Max. Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciation (Ap)

Part A
One Word Questions
Answer all questions
Each question carries 1 mark

1. **What is the effect of fat on gluten formation? (CO1 / A)**
   Ans: Fat inhibits gluten formation / shortens gluten strands
2. **What is the main function of carbohydrates? (CO1 / U)**
   Ans: Provide energy
3. **Name any two types of proteins present in cereals. (CO1 / K)**
   Ans: Glutenin, Gliadin *(or Prolamins, Glutelins)*
4. **Mention one microorganism causing sour curdling of milk. (CO2 / K)**
   Ans: *Lactobacillus*
5. **Name one milk product. (CO2 / K)**
   Ans: Curd *(or Paneer / Butter / Cheese)*
6. **Name the major protein in egg white. (CO2 / K)**
   Ans: Ovalbumin
7. **Name the principal flavonoid of citrus fruits. (CO3 / K)**
   Ans: Hesperidin
8. **What is sucrose? (CO4 / K)**
   Ans: Disaccharide (glucose + fructose)
9. **Give the temperature of soft ball stage of sugar cookery. (CO4 / K)**
   Ans: 112°C–115°C
10. **Expand FSSAI. (CO4 / K)**
    Ans: Food Safety and Standards Authority of India

(1×10 = 10 marks)

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

11. **Mention the different types of contamination in pulses. (CO1 / K)**
    Ans:

- Physical contamination
- Chemical contamination
- Microbial contamination
- Insect/pest contamination

12. **List four types of microorganisms important in food. (CO1 / K)**
    Ans:
    - Bacteria
    - Yeasts
    - Moulds
    - Viruses
13. **List the major functions of food. (CO1 / K)**
    Ans:
    - Body building
    - Energy giving
    - Protective/regulatory
14. **What are the preservation methods of eggs? (CO2 / K)**
    Ans:
    - Refrigeration
    - Oil coating
    - Lime water preservation
    - Freezing / dehydration
15. **Mention two methods of tenderising meat. (CO2 / K)**
    Ans:
    - Mechanical (pounding)
    - Enzymatic (papain/bromelain)
    - Marination
16. **Describe the fat in milk. (CO2 / U)**
    Ans:
    - Present as fat globules
    - Contains triglycerides
    - Emulsified by phospholipids and proteins
17. **List two methods of evaluating egg quality. (CO2 / K)**
    Ans:
    - Candling
    - Haugh unit
    - Shell examination
18. **Which are the preliminary preparations in vegetable cookery? (CO3 / A)**
    Ans:
    - Washing
    - Peeling
    - Cutting/slicing
    - Blanching (if needed)
19. **Write the post-harvest changes in fruits. (CO3 / U)**
    Ans:
    - Respiration
    - Ripening

- Moisture loss
- Textural softening

20. **State the nutritive value of green leafy vegetables. (CO3 / U)**

Ans:

- Rich in vitamins A, C, K
- High in iron, calcium
- Good fibre content
- Low calories

21. **Define food adulteration. (CO4 / K)**

Ans:

Intentional or unintentional addition/removal of substances that reduces food quality.

22. **State the nutritive value of fats. (CO4 / U)**

Ans:

- Energy dense (9 kcal/g)
- Essential fatty acids
- Carrier of fat-soluble vitamins

(2×10 = 20 marks)

Part C
Short Essay Questions
Answer any 5 questions
Each question carries 4 marks

23. **Explain the processing of pulses. (CO1 / A)**

Ans:

- Cleaning
- Grading
- Soaking
- Dehusking
- Drying
- Storage
- Optional: splitting (dal)

24. **Discuss the contamination and spoilage of cereal products. (CO1 / An)**

Ans:

- **Contamination:** insects, mites, dust, fungi
- **Spoilage:** mould growth, rancidity, insect infestation
- **Causes:** moisture, poor storage
- **Effects:** off-odour, discoloration, nutrient loss

25. **Describe the classification and nutritive value of poultry. (CO2 / An)**

Ans:

- **Classification:** broilers, layers, ducks, turkey
- **Nutritive value:** high-quality protein, B-vitamins, iron, low fat (depending on cut)

26. **Explain the spoilage and preservation of meat. (CO2 / U)**

Ans:

- **Spoilage:** microbial growth, slime formation, off-odour, discoloration
- **Preservation:** refrigeration, freezing, curing, smoking, canning, irradiation

27. **Discuss contamination and spoilage of fruits and vegetables. (CO3 / An)**
    Ans:
    - **Contamination:** pesticides, microbes, soil, handling
    - **Spoilage:** enzymatic browning, microbial decay, softening
    - **Causes:** moisture, temperature, mechanical injury
28. **Describe different types of rancidity and methods of prevention. (CO4 / U)**
    Ans:
    - **Types:**
        - Hydrolytic rancidity
        - Oxidative rancidity
    - **Prevention:** antioxidants, proper storage, avoiding light/heat, airtight containers
29. **Explain the changes of fats on heating. (CO4 / An)**
    Ans:
    - Melting
    - Smoking
    - Decomposition
    - Polymerisation
    - Formation of free radicals and trans fats

(4×5 = 20 marks)